

Predictor

Multiple predictors were tested on OpenRocket simulation data, including a simple Euler-step-based calculator, a simple vacuum-physics based function (F_1), and a derived drag and weathercock function (F_2). As the Euler method was inefficient and consumed too much compute time, a combination of F_1 and F_2 was used.

F_1 : Vacuum-Physics Based

$V_f^2 = V_0^2 + 2a\Delta y$	Start with this
$0 = V_0^2 + 2a\Delta y$	V_f is velocity at apogee, which is 0.
$\frac{-V_0^2}{2a} = \Delta y$	Solve for Δy
$H_{apogee} = H_0 + \frac{V_0^2}{2 a }$	Since this is for the coast phase, acceleration is negative (i.e. the minus signs cancel). Add current altitude to predict apogee.

Since this assumes a vacuum, it is more accurate at lower velocities where drag and weathercock are minimal.

F_2 : Derived, Accounts for Drag and Weathercocking

$H_f = y_0 + \frac{V_0^2}{2 a }$ $F_D = \frac{1}{2}C_D\rho v^2 A_{ref}$	Start with this
$a = -g - \frac{\frac{1}{2}C_D\rho v^2 A_{ref}}{m}$	Find acceleration due to gravity and drag.
$k = \frac{C_D\rho A_{ref}}{2m}$ $a = -g - kv^2 = -(g + kv^2)$	Since we only need v^2 to derive the function (the rest of the variables are either constant or can be retrieved from the sensors), we set the other variables equal to a constant k .
$H_f = y_0 + \frac{V_0^2}{2(g+kv^2)}$	Plug into F_1 .

$a = \frac{dv}{dt}$ $y = vt \Rightarrow dt = \frac{dy}{v}$ $\Rightarrow a = v \frac{dv}{dy}$ $-(g + kv^2) = v \frac{dv}{dy}$ $dy = \frac{-v}{g+kv^2}$	Prepare for integration by getting dy in terms of dv.
$\int_0^h dy = \int_{v_0}^0 \frac{-v}{g+kv^2} dv$ $u = g + kv^2, du = 2kv \cdot dv$ $u(v_0) = g + kv_0^2, u(0) = g$ $h = \int_{g+kv_0^2}^g \frac{1}{u} \cdot \frac{1}{-2k} du = \frac{1}{2k} \int_0^{g+kv_0^2} \frac{1}{u} du$ $h = \frac{1}{2k} [\ln g + kv^2 - \ln g]$ $h = \frac{1}{2k} \ln\left \frac{g+kv^2}{g}\right = \frac{1}{2k} \ln\left 1 + \frac{kv^2}{g}\right $	Integrate dv via a u-substitution. The bounds flip due to the negative sign.
$H_{apogee} = H_0 + \frac{1}{2k} \ln\left 1 + \frac{kv^2}{g \cdot \cos(\theta_{aoa})}\right \times \cos(\theta_{aoa})$	To approximate lost altitude due to angle of attack changes, we multiply both gravitational rotational acceleration and the entire function by the vertical component of acceleration.

Final Derived Apogee Predictor $H_{apogee} = H_0 + \frac{1}{2k} \ln\left 1 + \frac{kv^2}{g \cdot \cos(\theta_{aoa})}\right \times \cos(\theta_{aoa}), \text{ where } k = \frac{C_D \rho A_{ref}}{2m}$
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By inputting AoA, the last step accounts for all causes of AoA deviation, including weathercocking, manufacturing imperfections, and gravitational rotational acceleration.

Although this method yields generally strong results, higher velocities return more precise results, while lower velocities are somewhat worse than F_1 .

$$y_f = \frac{v^2}{2|a|} + y_0 \quad a = -g - \frac{\frac{1}{2} C_d \rho v^2 A}{m} \quad k = \frac{C_d \rho A}{2m} \quad a(v) = -g - kv^2$$

$$y_f = \frac{v^2}{2g + \frac{C_d \rho v^2 A}{m}} + y_0 \quad y_f = \frac{v^2}{2(g + kv^2)} + y_0 \quad a = \frac{dv}{dt} \quad dt = \frac{dv}{v} \quad a = v \frac{dv}{dy}$$

$$-(g + kv^2) = v \frac{dv}{dy} \quad dy = \frac{-v}{(g + kv^2)} dv \quad \int_0^h dy = \int_{v_0}^0 \frac{-v}{(g + kv^2)} dv$$

$$u = g + kv^2 \quad du = 2kv dv$$

$$u(v_0) = g + kv_0^2 \quad u(0) = g$$

$$h = \int_{g+kv_0^2}^g \frac{1}{u} \cdot \frac{1}{-2k} du$$

$$h = \frac{1}{-2k} \int_{g+kv_0^2}^g \frac{1}{u} du \quad h = \frac{1}{2k} \left[\ln|u| \right]_{g+kv_0^2}^g = \frac{1}{2k} \left[\ln|g| - \ln|g + kv_0^2| \right]$$

$$h = \frac{1}{2k} \ln \left| \frac{g}{g + kv_0^2} \right| + h_0$$

$$v_0 = v_{y0}$$